NICHOLAS W. LUNDGREN

Process Engineer | Semiconductor Materials | Data-Driven Characterization
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PROFESSIONAL SUMMARY

PhD candidate in physical chemistry with 5+ years of research in semiconductor thermal transport, materials characterization, and computational modeling. Experienced in designing experiments, analyzing large datasets, troubleshooting complex systems, and translating findings into clear technical documentation. Proven ability to mentor junior researchers, communicate across disciplines, and implement new analytical tools. Seeking to apply deep materials science knowledge and process optimization skills in a process engineering role.

CORE COMPETENCIES

- Process Development and Optimization
- Machine Learning and Scientific Computing
- Mentorship and Team Leadership
- Architecture-specific Compilation
- Data Analysis and Visualization (Python)
- HPC Workflows and Linux Systems
- Molecular and Lattice Dynamics
- Benchmarking and Troubleshooting
- Technical Documentation and Reporting
- Cross-functional Communication
- Raman and Brillouin Spectroscopy
- Python, Fortran, Git, LaTeX

EXPERIENCE

Graduate Researcher, Department of Chemistry, UC Davis *Sep 2020 to Present*

- Designed and executed thermal transport experiments on semiconductor and amorphous materials using molecular dynamics, lattice dynamics, and machine learning potentials, analyzing results to inform materials design.
- Performed characterization of low-symmetry and disordered material systems, interpreting bulk structure and thin film properties through computational methods.
- Diagnosed and resolved complex simulation failures across HPC platforms, including debugging software builds, compiler issues, and architecture-specific bottlenecks.
- Measured and interpreted thermal conductivity, vibrational spectra, and phonon transport properties in thin-film silicon, amorphous carbon, and 2D allotropes (silicene).
- Co-developed the open-source kALDo thermal transport code, implementing new features, automating documentation pipelines, and evaluating third-party tools for integration.
- Trained machine learning interatomic potentials (NEP) to enable large-scale simulations previously inaccessible through traditional methods.

Teaching Assistant, Department of Chemistry, UC Davis *2020 to 2026*

- Communicated complex technical material to diverse audiences across general chemistry, thermodynamics, spectroscopy, quantum chemistry, kinetics, and organic chemistry courses.
- Mentored an undergraduate researcher through experimental design and data interpretation, serving as a technical resource for junior colleagues.
- Nominated for Best Teaching Assistant for excellence in instructional delivery and student engagement.

EDUCATION

PhD in Chemistry (Physical Chemistry), University of California, Davis *Expected June 2026*
Focus: Thermal transport in semiconductor and disordered materials; scientific software development; ML for atomistic modeling. GPA: 3.75

BS in Chemical Physics, Highest Honors, University of California, Davis *June 2020*
Thesis: The Effects of Germanium Concentration on the Thermal Transport of Amorphous Silicon. GPA: 3.61

SELECTED PUBLICATIONS

- Layer-Stacking and Strain Effects on Heat Transport in Silicene (2026, submitted)
- kALDo 2.0: Thermal Transport from Machine Learning Potentials, Computer Physics Communications (2026, accepted)
- Mode Localization and Suppressed Heat Transport, Physical Review B (2021)
- Efficient Anharmonic Lattice Dynamic Calculations, Journal of Applied Physics (2020)

TECHNICAL TOOLS

Software: LAMMPS, GPUMD, DLPOLY, ASE, kALDo, TensorFlow, Kokkos, VMD, Inkscape, Git, LaTeX | Languages: Python, Fortran, C++, JavaScript, HTML/CSS | Spoken: English (native), Portuguese (fluent)